

PIT Embedding Report

Embedding Quality Report: PIT-1B Base Model

To: Prof. [Supervisor] **From:** Yutong Yan **Date:** August 20, 2026 **Re:** Hidden-state quality issue discovered during earnings call embedding pipeline development

Summary

While building a point-in-time embedding pipeline for earnings call transcripts, we tested two model families: **PIT** (Diamegs) and **DatedGPT**. Both provide annual checkpoints trained on data up to a specific year, enabling look-ahead-free embeddings.

We found that the PIT-1B base model produces embeddings with a structural quality issue that makes them unsuitable for direct use in financial applications. DatedGPT does not exhibit this issue. We document the finding below and suggest potential remedies.

Test Setup

We embedded the **same 200 quarterly earnings call transcripts** (2014 Q1, US firms) using both models. Each transcript was truncated to 512 tokens and embedded via mean pooling over the last hidden layer.

Results

Metric	DatedGPT-2013-base	PIT-1B-201312
Hidden dimension	2,048	1,536

Metric	DatedGPT-2013-base	PIT-1B-201312
Embedding mean	-0.008	-0.079
Embedding std	0.80	14.62
Max absolute value	23.7	647.0
Dimensions with abs > 10	4 (0.2%)	1,449 (94%)
L2 norm (mean)	36.2	566.3

The issue: dominant outlier dimensions

In the PIT-1B base model, **two hidden dimensions carry extremely large, nearly constant values across all documents**:

- Dimension 1531: mean = 406.5, std = 76.0
- Dimension 1352: mean = -217.4, std = 44.7

These dimensions account for a disproportionate share of the embedding vector's magnitude. Because their variance is low relative to their mean (std/mean ratio of ~0.19), they carry very little information about differences between transcripts. In effect, they act as large bias terms that overshadow the meaningful variation in other dimensions.

Why this matters for financial applications

When using embeddings as features (e.g., predicting abnormal returns, constructing text-based surprise measures), the outlier dimensions would:

1. **Dominate any distance or similarity metric** (cosine similarity, L2 distance), making all documents appear nearly identical
2. **Overwhelm regression coefficients**, since a regularized model would allocate most weight to explaining the large-magnitude dimensions rather than the informative ones
3. **Reduce the effective dimensionality** of the representation from 1,536 to a much smaller number

DatedGPT's embeddings do not have this problem. Its largest dimension (mean = 20.1) is 20x smaller than PIT's, and 99.8% of dimensions stay below an absolute value of 10.

Possible Explanations

This phenomenon has been documented in the transformer literature under terms like “massive activations” or “outlier features.”

Contributing factors may include:

- The depth of the network (PIT-1B has 52 layers vs DatedGPT’s 24), which allows more accumulation through residual connections
- Differences in initialization or normalization placement between the architectures
- The language modeling objective does not directly constrain hidden-state magnitudes, since the LM head projection can compensate

We have not done a layer-by-layer analysis to pinpoint the exact cause, and this would be a worthwhile investigation for the PIT team.

Potential Remedies

Approach	Feasibility	Effectiveness
Per-dimension standardization (zero mean, unit variance)	Easy, post-hoc	Good, but may amplify noise in low-variance dims
Use PIT-4B-FT (fine-tuned variant)	Easy if available	Reportedly better, but needs verification
Intermediate-layer embeddings (e.g., layer 26 instead of 51)	Easy to test	Outliers may be less severe in earlier layers
Architectural fix (e.g., layer norm after last block, hidden-state regularization during training)	Requires retraining	Would resolve the root cause
Use DatedGPT instead	Immediate	Avoids the issue entirely

Current Decision

For our earnings call embedding pipeline, we are proceeding with **DatedGPT-base** as the default model. It provides clean embeddings, standard HuggingFace compatibility, and the same annual point-in-time checkpoints (2013-2024).

We remain open to switching to PIT if the outlier issue is addressed (e.g., through a future model release or a validated normalization procedure).

Appendix: Raw Embedding Statistics

DatedGPT-2013-base, 10 random samples from 200 earnings calls

```
Sample [4216]: mean=-0.062, std=15.08, min=-226.7, max=439.1  <-  
PIT
```

```
Sample [4216]: mean=-0.008, std=0.80, min=-20.3, max=23.7   <-  
DatedGPT (typical)
```

The PIT sample's max (439) is 18x larger than DatedGPT's max (23.7), and both come from the same transcript.

Top outlier dimensions in PIT-1B-201312 (200 earnings calls)

```
dim 1531: abs_max=647.0, mean=406.5, std=76.0  (std/mean = 0.19)  
dim 1352: abs_max=379.0, mean=-217.4, std=44.7  (std/mean = 0.21)  
dim 1081: abs_max=139.0, mean=21.9,  std=17.8  (std/mean = 0.81)
```

Dimensions 1531 and 1352 are nearly constant across documents (low std/mean ratio), confirming they carry minimal discriminative information.